

SSL Certificate Update Runbook

Nginx lb-server container - ai-training.cck.rnu.tn

Target domain	ai-training.cck.rnu.tn
New certificate validity	15 Apr 2026 14:59:43 GMT to 14 Jul 2026 14:59:42 GMT
Remote VM	data-node01 / 193.95.31.101
Nginx runtime	Docker container: lb-server, container id observed: e2e1e2ed5f0a
Active mounted host path	/opt/lb/conf/vhosts/ai/fullchain1.pem and /opt/lb/conf/vhosts/ai/privkey1.pem
Container path	/usr/local/NSP/etc/nginx/vhosts/ai/fullchain1.pem and /usr/local/NSP/etc/nginx/vhosts/ai/privkey1.pem

Result: The certificate was replaced on the active mounted Nginx path, ownership was corrected for the container user uid=10010/gid=10010, Nginx syntax test passed, and reload_nginx.sh reported success.

1. Local certificate verification

The new certificate files were located on the local workstation at:

```
~/Downloads/cert-ai-training
```

The working files were:

- cert.pem
- chain.pem
- fullchain.pem
- privkey.pem
- README

Verify the certificate subject, issuer and validity period:

```
cd ~/Downloads/cert-ai-training
openssl x509 -in fullchain.pem -noout -subject -issuer -dates
```

Expected output from the successful run:

```
subject=CN = ai-training.cck.rnu.tn
issuer=C = US, O = Let's Encrypt, CN = E7
notBefore=Apr 15 14:59:43 2026 GMT
notAfter=Jul 14 14:59:42 2026 GMT
```

Verify that the certificate and private key match:

```
echo "CERT HASH:"
openssl x509 -in fullchain.pem -noout -pubkey | openssl pkey -pubin -outform der | openssl dgst -sha256

echo "KEY HASH:"
openssl pkey -in privkey.pem -pubout -outform der | openssl dgst -sha256
```

Expected hash from the successful run:

```
8c819859e78a4239236ff1a53bb7db5df9b0ba3e2f6d202bf2393e6ecd65abf3
```

2. Upload the new certificate to the remote VM

The first attempt to upload directly to /paas/SSL-AI failed because the real directory was /home/paas/SSL-AI. The working upload was to the paas user home directory:

```
cd ~/Downloads/cert-ai-training
```

```
scp fullchain.pem privkey.pem cert.pem chain.pem README paas@193.95.31.101:~/
```

After the upload, the new files existed on the remote VM as:

```
/home/paas/fullchain.pem
/home/paas/privkey.pem
/home/paas/cert.pem
/home/paas/chain.pem
/home/paas/README
```

3. Confirm remote paths and prepare local copies

On the remote VM, confirm the paas home directory and certificate files:

```
cd /home/paas
pwd
ls -lh
```

The following files were found and used for consistency copies:

```
/home/paas/fullchain.pem
/home/paas/privkey.pem
/home/paas/fullchain1.pem
/home/paas/privkey1.pem
/home/paas/SSL-AI
```

Create a backup of the old /home/paas certificate pair:

```
cd /home/paas
STAMP="$(date +%Y%m%d_%H%M%S)"
mkdir -p /home/paas/SSL-AI/backup-$STAMP
cp -a /home/paas/fullchain1.pem /home/paas/SSL-AI/backup-$STAMP/
cp -a /home/paas/privkey1.pem /home/paas/SSL-AI/backup-$STAMP/
echo "Backup created: /home/paas/SSL-AI/backup-$STAMP"
```

Because cp was aliased to interactive mode, use install instead of cp -f to overwrite safely:

```
cd /home/paas
install -o paas -g paas -m 600 /home/paas/fullchain.pem /home/paas/fullchain1.pem
install -o paas -g paas -m 600 /home/paas/privkey.pem /home/paas/privkey1.pem
```

Verify the consistency pair:

```
openssl x509 -in /home/paas/fullchain1.pem -noout -subject -issuer -dates

openssl x509 -in /home/paas/fullchain1.pem -noout -pubkey | openssl pkey -pubin -outform der | openssl dgst -sha256

openssl pkey -in /home/paas/privkey1.pem -pubout -outform der | openssl dgst -sha256
```

4. Identify the active Nginx container and mounted configuration

The host nginx -T command did not show the active SSL configuration because Nginx was running inside a Docker container named lb-server.

Find the lb-server container:

```
docker ps | grep -i "lb-server\|nginx"
```

Observed active container:

```
e2e1e2ed5f0a registry.fusionstage.local:20202/gde/lb-server:25.1.510000003 ... k8s_lb-service_lb-service-0_kube-
system_...
```

Set the container ID:

```
CID="e2e1e2ed5f0a"
echo "$CID"
```

Inspect container mounts:

```
docker inspect "$CID" --format '{{range .Mounts}}{{println .Source " -> " .Destination}}{{end}}'
```

Important mount discovered:

```
/opt/lb/conf -> /usr/local/NSP/etc/nginx
```

Search active SSL references inside the container:

```
docker exec "$CID" bash -lc '
/usr/local/NSP/sbin/nginx -T 2>&1 | grep -nE "server_name|listen .*443|ssl_certificate|ssl_certificate_key" || true
'
```

Active configuration discovered:

```
listen 172.16.82.18:443 ssl;
server_name 193.95.31.101 www.ai-training.cck.rnu.tn training.cck.rnu.tn;
ssl_certificate vhosts/ai/fullchain1.pem;
ssl_certificate_key vhosts/ai/privkey1.pem;
```

Important finding: The active SSL files were not /home/paas/fullchain1.pem directly. They were the files under the mounted Nginx configuration path: /opt/lb/conf/vhosts/ai/fullchain1.pem and /opt/lb/conf/vhosts/ai/privkey1.pem.

5. Backup and replace the real active Nginx certificate

Backup the real active files before replacing them:

```
cd /home/paas
STAMP="$(date +%Y%m%d_%H%M%S)"
mkdir -p /home/paas/SSL-AI/backup-nginx-active-$STAMP
cp -a /opt/lb/conf/vhosts/ai/fullchain1.pem /home/paas/SSL-AI/backup-nginx-active-$STAMP/
cp -a /opt/lb/conf/vhosts/ai/privkey1.pem /home/paas/SSL-AI/backup-nginx-active-$STAMP/
echo "Backup created: /home/paas/SSL-AI/backup-nginx-active-$STAMP"
```

Replace the real active Nginx certificate pair:

```
install -m 600 /home/paas/fullchain.pem /opt/lb/conf/vhosts/ai/fullchain1.pem
install -m 600 /home/paas/privkey.pem /opt/lb/conf/vhosts/ai/privkey1.pem
```

Verify the active certificate on the host:

```
openssl x509 -in /opt/lb/conf/vhosts/ai/fullchain1.pem -noout -subject -issuer -dates
```

Expected successful output:

```
subject=CN = ai-training.cck.rnu.tn
issuer=C = US, O = Let's Encrypt, CN = E7
notBefore=Apr 15 14:59:43 2026 GMT
notAfter=Jul 14 14:59:42 2026 GMT
```

Verify the active certificate and key match:

```
echo "CERT HASH:"
openssl x509 -in /opt/lb/conf/vhosts/ai/fullchain1.pem -noout -pubkey | openssl pkey -pubin -outform der | openssl dgst -sha256

echo "KEY HASH:"
openssl pkey -in /opt/lb/conf/vhosts/ai/privkey1.pem -pubout -outform der | openssl dgst -sha256
```

Expected hash for both outputs:

```
8c819859e78a4239236ff1a53bb7db5df9b0ba3e2f6d202bf2393e6ecd65abf3
```

6. Fix permissions for the lb-server container user

The Nginx test initially failed because the key was root:root with mode 600, while the container process runs as user lb.

Confirm container user and Nginx processes:

```
CID="e2e1e2ed5f0a"
docker exec "$CID" bash -lc '
echo "Container current user:"
id

echo
```

```
ps -ef | grep "[n]ginx"

echo

ls -ln /usr/local/NSP/etc/nginx/vhosts/ai/fullchain1.pem      /usr/local/NSP/etc/nginx/vhosts/ai/privkey1.pem
,
```

Observed container user:

```
uid=10010(lb) gid=10010(lb) groups=10010(lb)
```

Apply correct ownership and secure permissions on the host-mounted active files:

```
CID="e2e1e2ed5f0a"
CUID="$(docker exec "$CID" id -u)"
CGID="$(docker exec "$CID" id -g)"

echo "Container UID=$CUID GID=$CGID"

chown "$CUID:$CGID" /opt/lb/conf/vhosts/ai/fullchain1.pem /opt/lb/conf/vhosts/ai/privkey1.pem
chmod 644 /opt/lb/conf/vhosts/ai/fullchain1.pem
chmod 600 /opt/lb/conf/vhosts/ai/privkey1.pem

ls -lh /opt/lb/conf/vhosts/ai/fullchain1.pem /opt/lb/conf/vhosts/ai/privkey1.pem
```

Expected ownership after the successful fix:

```
-rw-r--r-- 1 10010 10010 2.9K ... /opt/lb/conf/vhosts/ai/fullchain1.pem
-rw----- 1 10010 10010 227 ... /opt/lb/conf/vhosts/ai/privkey1.pem
```

7. Test and reload Nginx

Test the Nginx configuration inside the lb-server container:

```
CID="e2e1e2ed5f0a"
docker exec "$CID" bash -lc '/usr/local/NSP/sbin/nginx -t'
```

Expected successful output:

```
nginx: the configuration file /usr/local/NSP/etc/nginx/nginx.conf syntax is ok
nginx: configuration file /usr/local/NSP/etc/nginx/nginx.conf test is successful
```

Reload Nginx using the platform script:

```
cd /home/paas
./reload_nginx.sh
```

Expected successful output:

```
[INFO]    reload_nginx ...
[INFO]    reload_nginx Successful.
```

8. Final validation

After the reload, validate what the public endpoint is serving:

```
echo | openssl s_client -connect ai-training.cck.rnu.tn:443 -servername ai-training.cck.rnu.tn 2>/dev/null | openssl
x509 -noout -subject -issuer -dates
```

Expected result:

```
subject=CN = ai-training.cck.rnu.tn
issuer=C = US, O = Let's Encrypt, CN = E7
notBefore=Apr 15 14:59:43 2026 GMT
notAfter=Jul 14 14:59:42 2026 GMT
```

Optional local client check:

```
curl -Iv https://ai-training.cck.rnu.tn
```

9. Rollback procedure

Use this only if Nginx fails after replacement and you need to restore the previous active pair.

```
# Replace BACKUP_DIR with the exact backup folder created earlier.
BACKUP_DIR="/home/paas/SSL-AI/backup-nginx-active-YYYYMMDD_HHMMSS"
CID="e2e1e2ed5f0a"
CUID="$(docker exec "$CID" id -u)"
CGID="$(docker exec "$CID" id -g)"

install -m 600 "$BACKUP_DIR/fullchain1.pem" /opt/lb/conf/vhosts/ai/fullchain1.pem
install -m 600 "$BACKUP_DIR/privkey1.pem" /opt/lb/conf/vhosts/ai/privkey1.pem

chown "$CUID:$CGID" /opt/lb/conf/vhosts/ai/fullchain1.pem /opt/lb/conf/vhosts/ai/privkey1.pem
chmod 644 /opt/lb/conf/vhosts/ai/fullchain1.pem
chmod 600 /opt/lb/conf/vhosts/ai/privkey1.pem

docker exec "$CID" bash -lc '/usr/local/NSP/sbin/nginx -t'
cd /home/paas
./reload_nginx.sh
```

10. Troubleshooting notes from this update

Issue	Working resolution
Do not type Nginx directives in the shell	Lines such as <code>ssl_certificate ...</code> ; and <code>ssl_certificate_key ...</code> ; are configuration directives, not shell commands.
Do not rely only on <code>nginx -T</code> on the host	The active Nginx was inside the lb-server Docker container, not the host Nginx service.
<code>cp -f</code> may still prompt if <code>cp</code> is aliased	On this server, <code>cp</code> behaved interactively. The working approach was <code>install</code> , not <code>cp -f</code> .
Use the mounted host path for persistence	The container path <code>/usr/local/NSP/etc/nginx</code> is mounted from <code>/opt/lb/conf</code> on the host. Updating <code>/opt/lb/conf/vhosts/ai</code> makes the change persistent.
Private key permissions matter	The container runs as <code>uid=10010/gid=10010</code> . The private key must be readable by that user; 600 is safe only if ownership is also 10010:10010.
Ignore inactive certificate names	<code>certificate.pem</code> , <code>private.key</code> , <code>server.pem</code> , <code>server.key</code> and <code>/ssl/ai</code> were not active for this site according to the real Nginx configuration.

11. Condensed working command sequence

This is the final minimal sequence after the new files have already been uploaded to `/home/paas`.

```
# Remote VM - run as root
cd /home/paas
CID="e2e1e2ed5f0a"

# 1) Back up active Nginx cert pair
STAMP="$(date +%Y%m%d_%H%M%S)"
mkdir -p /home/paas/SSL-AI/backup-nginx-active-$STAMP
cp -a /opt/lb/conf/vhosts/ai/fullchain1.pem /home/paas/SSL-AI/backup-nginx-active-$STAMP/
cp -a /opt/lb/conf/vhosts/ai/privkey1.pem /home/paas/SSL-AI/backup-nginx-active-$STAMP/

# 2) Replace active certificate and key
install -m 600 /home/paas/fullchain.pem /opt/lb/conf/vhosts/ai/fullchain1.pem
install -m 600 /home/paas/privkey.pem /opt/lb/conf/vhosts/ai/privkey1.pem
```

```
# 3) Set ownership for the container user
CUID="$(docker exec "$CID" id -u)"
CGID="$(docker exec "$CID" id -g)"
chown "$CUID:$CGID" /opt/lb/conf/vhosts/ai/fullchain1.pem /opt/lb/conf/vhosts/ai/privkey1.pem
chmod 644 /opt/lb/conf/vhosts/ai/fullchain1.pem
chmod 600 /opt/lb/conf/vhosts/ai/privkey1.pem

# 4) Verify cert and key
openssl x509 -in /opt/lb/conf/vhosts/ai/fullchain1.pem -noout -subject -issuer -dates
openssl x509 -in /opt/lb/conf/vhosts/ai/fullchain1.pem -noout -pubkey | openssl pkey -pubin -outform der | openssl dgst
-sha256
openssl pkey -in /opt/lb/conf/vhosts/ai/privkey1.pem -pubout -outform der | openssl dgst -sha256

# 5) Test and reload Nginx
docker exec "$CID" bash -lc '/usr/local/NSP/sbin/nginx -t'
cd /home/paas
./reload_nginx.sh

# 6) External verification
echo | openssl s_client -connect ai-training.cck.rnu.tn:443 -servername ai-training.cck.rnu.tn 2>/dev/null | openssl
x509 -noout -subject -issuer -dates
```

Operational status: The successful reload message confirms the running lb-server Nginx accepted the new certificate configuration. Always keep the generated backup folder until the new certificate has been validated from an external client.